



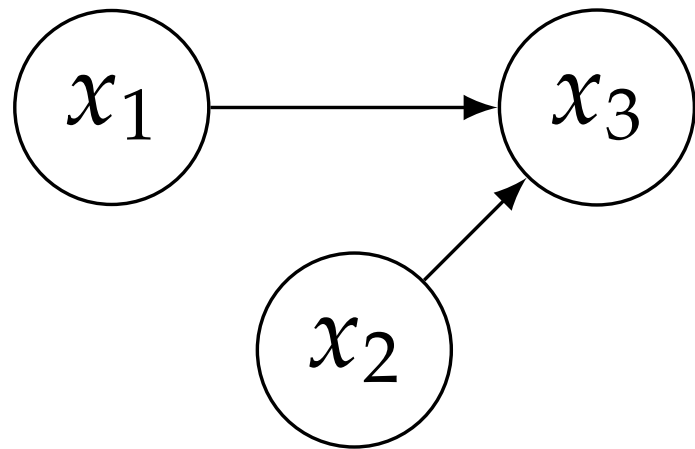
DATA SCIENCE &
ARTIFICIAL INTELLIGENCE

Probabilistic Machine Learning

Expectation Maximisation

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Learning Bayesian Networks



$$p(x_1, x_2, x_3) = p(x_3|x_1, x_2)p(x_1)p(x_2)$$

x_j binary

$$p(x_3 = 1|x_1 = 0, x_2 = 0) = \theta_{00}$$

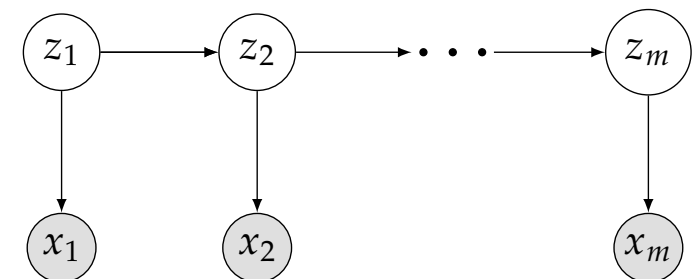
$$p(x_3 = 1|x_1 = 0, x_2 = 1) = \theta_{01}$$

$$p(x_3 = 1|x_1 = 1, x_2 = 0) = \theta_{10}$$

$$p(x_3 = 1|x_1 = 1, x_2 = 1) = \theta_{11}$$

$$\theta_{ij} = \frac{\#(x_1 = i, x_2 = j, x_3 = 1)}{\#(x_1 = i, x_2 = j)},$$

what if some x_j is not observed?



Latent variables

x observed, z latent

$$p(x|\theta) = \sum_z p(x, z|\theta)$$

$$\underline{x} = x_1, \dots, x_n$$

$$\underline{z} = z_1, \dots, z_n$$

$$p(\underline{x}, \underline{z}|\theta) = \prod_n p(x_n, z_n|\theta)$$

$$\log p(\underline{x}, \underline{z}|\theta) = \sum_n \log p(x_n, z_n|\theta)$$

$$\log p(\underline{x}|\theta) \neq \sum_n \log p(x_n|\theta)$$

Evidence Lower Bound

$p(x, z)$ observations $x_1, \dots, x_n$ and latent states $z_1, \dots, z_n$

$$p(x|\theta) = \sum_z p(x, z|\theta)$$

$$\operatorname{argmax}_{\theta} p(\underline{x}|\theta)$$

$$p(x, z|\theta) = p(x|\theta)p(z|x, \theta)$$

introduce a variational approximation $q(z)$ of $p(z | x, \theta)$

Evidence Lower Bound

$$KL[q||p] = KL[q(z)||p(z|x, \theta)] = \mathbb{E}_{q(z)} \left[-\log \frac{p(z|x, \theta)}{q(z)} \right] = - \sum_z q(z) \log \frac{p(z|x, \theta)}{q(z)}$$

$$- \sum_z q(z) \log \frac{p(z|x, \theta)}{q(z)} = - \sum_z q(z) \left[\log \frac{p(z|x, \theta)}{q(z)} + \log p(x|\theta) - \log p(x|\theta) \right]$$

$$KL[q(z)||p(z|x, \theta)] = - \sum_z q(z) \log \frac{p(x, z|\theta)}{q(z)} + \log p(x|\theta)$$

$$\mathcal{L}(q, \theta) = \sum_z q(z) \log \frac{p(x, z|\theta)}{q(z)}$$

is the Evidence Lower Bound (also called ELBO).

Evidence Lower Bound

$$\log p(x | \theta) = \mathcal{L}(q, \theta) + KL[q(z) || p(z | x, \theta)]$$

$$\mathcal{L}(q, \theta) = \sum_z q(z) \log \frac{p(x, z | \theta)}{q(z)}$$

Since $KL[q | p] \geq 0$ then $\mathcal{L}(q, \theta) \leq \log(p(x | \theta))$

E-step

$$\mathcal{L}(q, \theta) = \mathbb{E}_q \left[\log \frac{p(x, z | \theta)}{q(z)} \right] = \mathbb{E}_q [\log p(x, z | \theta)] + \mathbb{E}_q [-\log q(z)]$$

$$\mathcal{L}(q, \theta) = \log p(\underline{x} | \theta) - KL[q(z) || p(z | \underline{x}, \theta)]$$

fix θ to θ_{old}

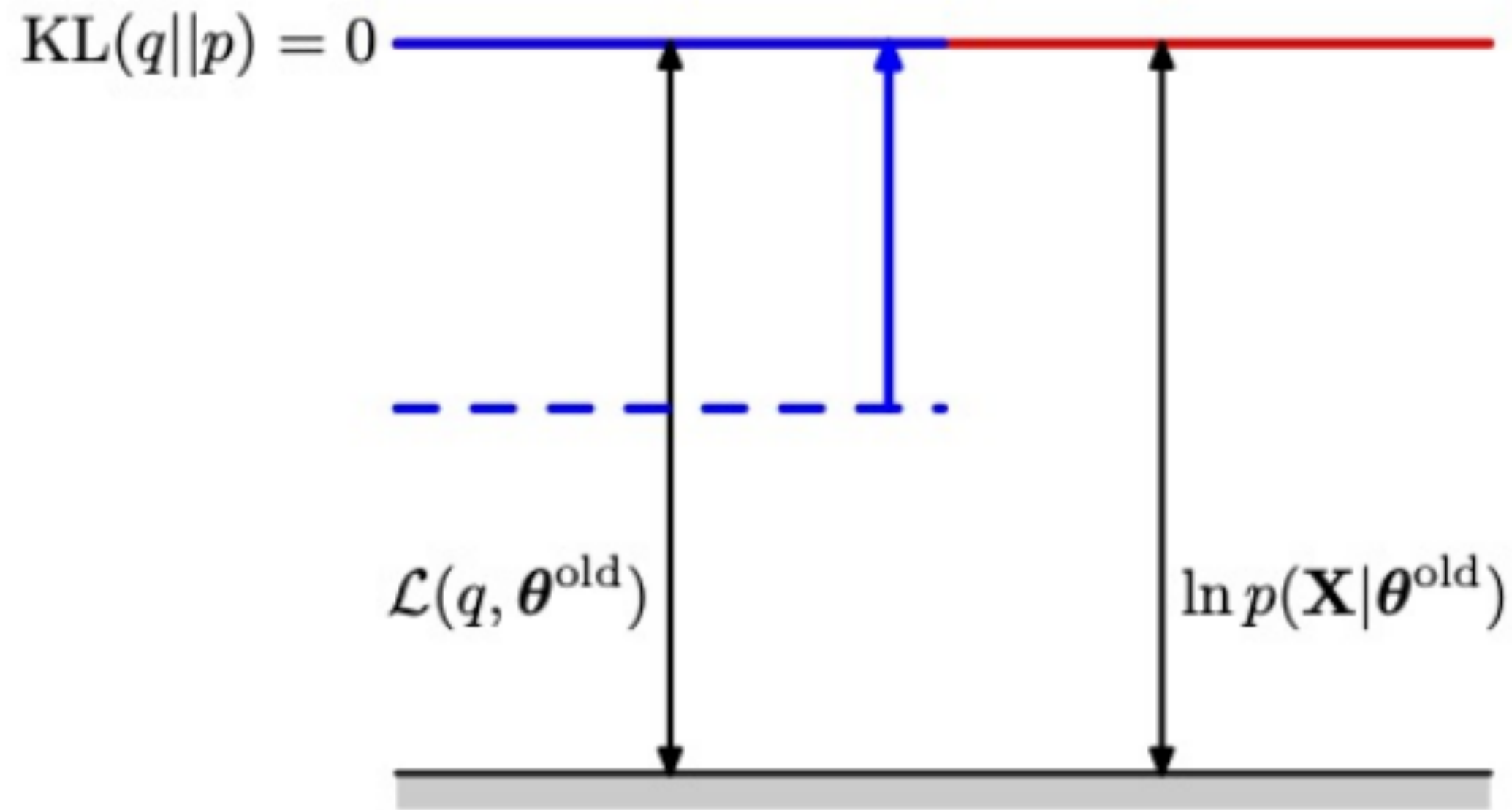
$\mathcal{L}(q, \theta_{old})$ is maximum whenever $KL = 0$

$$q_{max} = p(\underline{z} | \underline{x}, \theta_{OLD})$$

$$p(\underline{z} | \underline{x}, \theta) = \prod_{i=1}^N p(z_i | x_i, \theta) \quad \text{if } x_1, \dots, x_n \text{ are i.i.d}$$

Then we compute $\mathbb{E}_{q_{new}} [\log p(\underline{x}, z | \theta)]$

E-step



M-step

maximize $\mathcal{L}(q, \theta)$ keeping q fixed to $p(\underline{z}|\underline{x}, \theta_{OLD})$ i.e. maximize

$$\mathbb{E}_{q_{new}}[\log p(\underline{z}, \underline{x}|\theta)]$$

$$\theta_{new} = \operatorname{argmax}_{\theta} \mathbb{E}_{q_{new}}[\log p(\underline{x}, \underline{z}|\theta)]$$

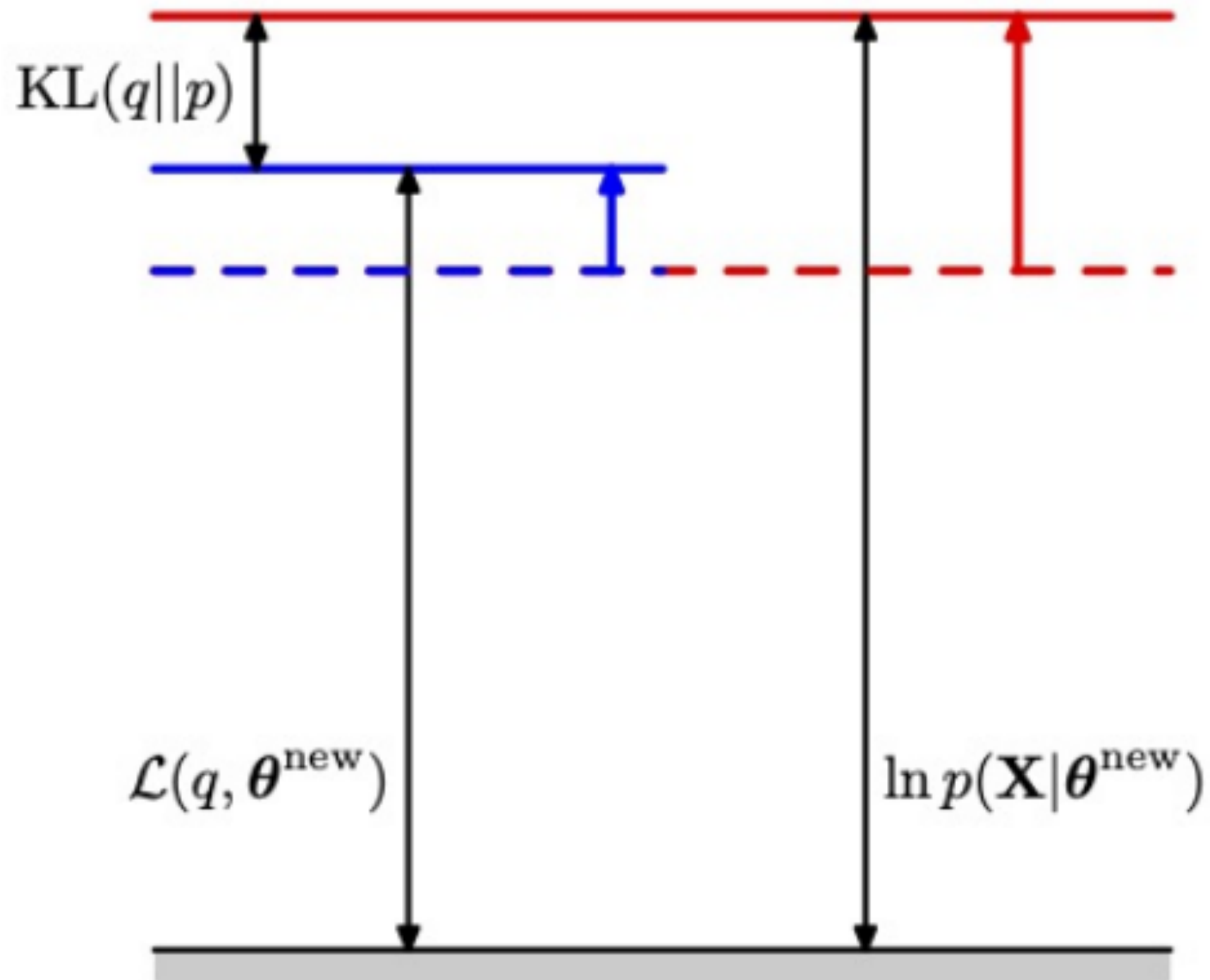
by computing the gradient and setting it to zero, or numerically

$$\text{repeat until } \|\theta_{OLD} - \theta_{NEW}\| < \varepsilon$$

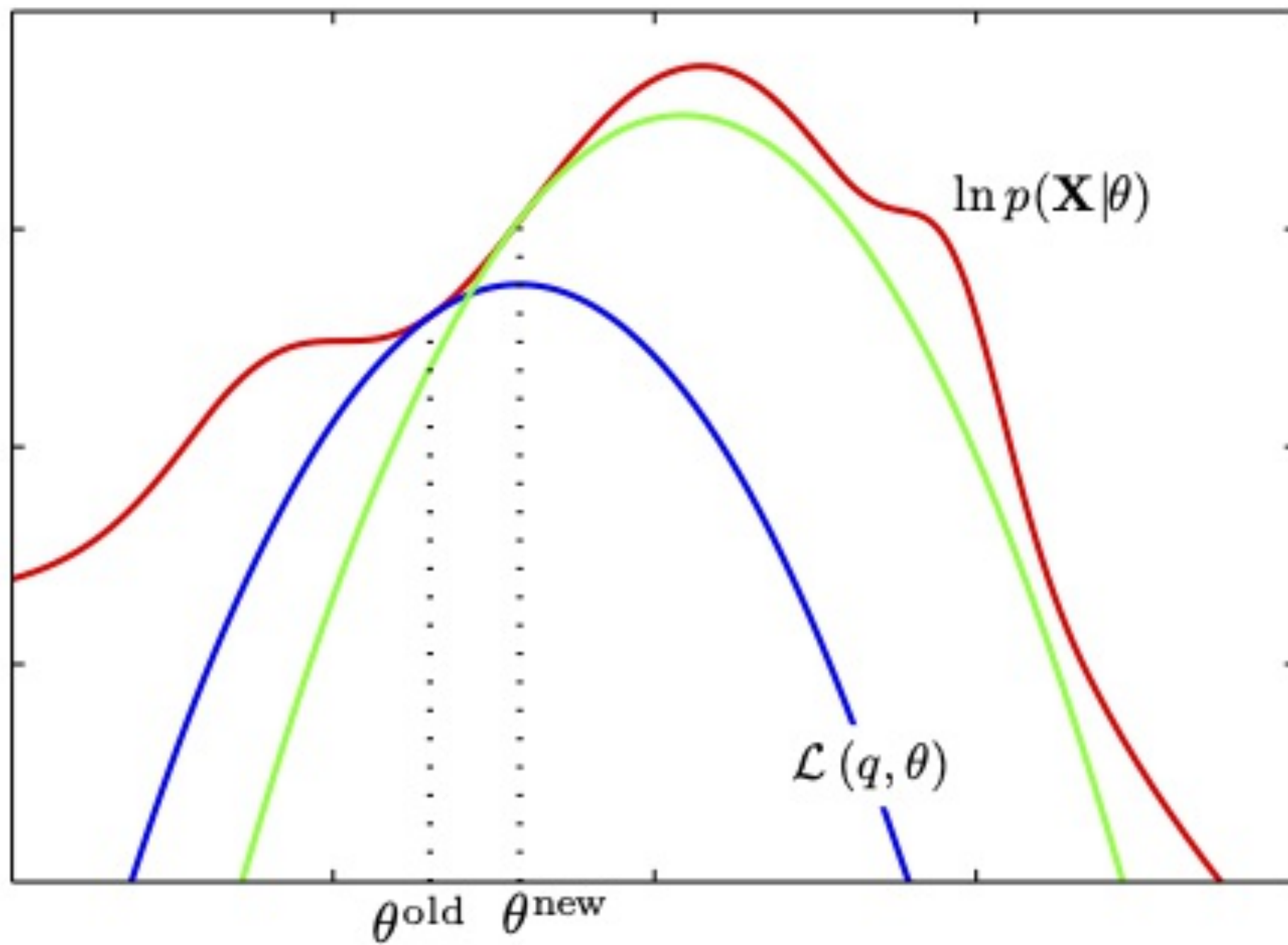
Remark In general the EM algorithm converges to a **local** optimum of $\log p(\underline{x}|\theta)$.

Remark If $p(\underline{z}|\underline{x}, \theta)$ is not easily computable, we can use a q that is an approximation. In this case EM iterations will reduce $\text{KL}(q||p)$ but it will not be set to zero, hence convergence is not guaranteed.

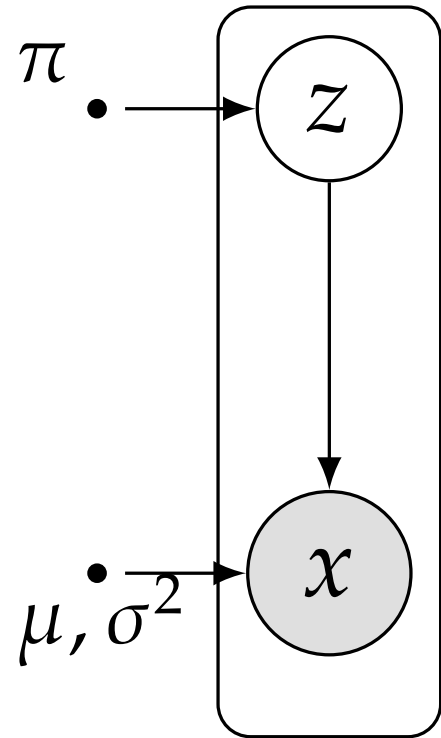
M-step



M-step



Mixture of Gaussians



$$z_1, \dots, z_k, z_j \in \{0, 1\}, \sum_j z_j = 1$$

$$p(x, z | \theta) = \prod_{j=1}^K \pi_j^{z_j} \cdot \mathcal{N}(x | \mu_j, \sigma_j^2)^{z_j}$$

N observations $\underline{x} = x_1, \dots, x_n$

$$p(x | \theta) = \sum_{j=1}^K \pi_j \mathcal{N}(x | \mu_j, \sigma_j^2)$$

$$p(z | \theta) = \prod_j \pi_j^{z_j}$$

$$p(z | \underline{x}, \theta) \propto \prod_{n=1}^N \prod_{j=1}^K \pi_j^{z_{nj}} \mathcal{N}(x_n | \mu_j, \sigma_j^2)^{z_{nj}}$$

Mixtures of Gaussians

E-step - computing expectations

$$\mathbb{E}_{p(z|x)}[z_{nj}] = p(z = j|x_n, \theta) = \frac{\pi_j \mathcal{N}(x_n|\mu_j, \sigma_j^2)}{\sum_i \pi_i \mathcal{N}(x_n|\mu_i, \sigma_i^2)} := \gamma(z_{nj})$$

$$\log p(\underline{x}, \underline{z}|\theta) = \sum_{n=1}^N \sum_{j=1}^K z_{nj} [\log \pi_j + \log \mathcal{N}(x_n|\mu_j, \sigma_j^2)]$$

$$\mathbb{E}_{p(\underline{z}|\underline{x}, \theta)}[\log p(\underline{x}, \underline{z}|\theta)] = \sum_{n=1}^N \sum_{j=1}^K \mathbb{E}[z_{nj}] [\log \pi_j + \log \mathcal{N}(x_n|\mu_j, \sigma_j^2)]$$

Mixtures of Gaussians

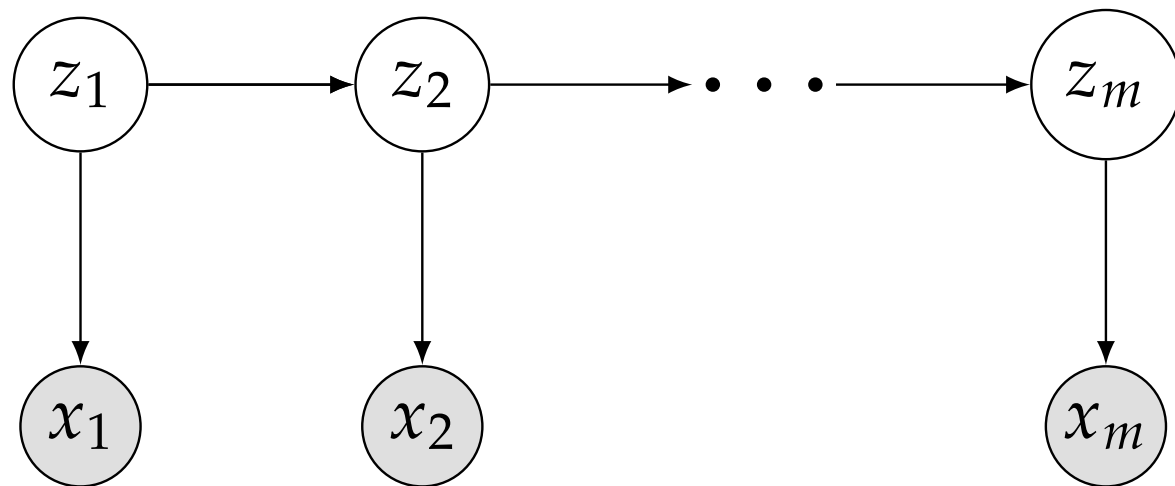
M-step - maximising parameters (compute gradients and set = 0)

$$\mu_j^{new} = \frac{1}{N_j} \sum_n \gamma(z_{nj}) x_n$$

$$\Sigma_j^{new} = \frac{1}{N_j} \sum_n \gamma(z_{nj}) (x_n - \mu_j^{new})^T (x_n - \mu_j^{new})$$

$$\pi_j^{new} = \frac{N_j}{N}, \quad N_j = \sum_{n=1}^N \gamma(z_{nj})$$

Hidden Markov Models (Baum Welch)



$$p(z_1 = i) = \pi_i$$

$$p(z_i = j | z_{i-1} = k) = A_{kj}$$

$$p(x_i | z_i = k) = p(x_i | \phi_k)$$

$$\theta = (\pi, A, \phi)$$

$\underline{x} = x^1, \dots, x^N$ where each $x^i = (x_1^i, \dots, x_M^i)$

$q^n(z) = p(z | x^n, \theta_{OLD}) \forall n$ by message passing

$$E(\theta) = \sum_{n=1}^N \left[\sum_{k=1}^K q^n(z_{1k}) \ln \pi_k + \sum_{i=2}^M \sum_{j,k=1}^K q^n(z_{(i-1)j}, z_{ik}) \ln A_{jk} \right. \\ \left. + \sum_{i=1}^M \sum_{k=1}^K q^n(z_{ik}) \ln p(x_i^n | \phi_k) \right] \Rightarrow \pi_k = \frac{\sum_n q^n(z_{1k})}{\sum_i \sum_n q^n(z_{1i})}.$$

